



A Rolls-Royce solution

designation

circuit-diagram "control-panel" BR4000ECU9 DEIF AGC-4 A10

project number

XZ54934000034 (ZNG00040842)

material number

see table-item

serial number

type-of-system

[illegible]

INFORMATION TO SWITCHGEAR ASSEMBLY

	generally	auxiliary-drives	voltage sensing busbar / mains
type-of-electrical-system	TN-S		
rated operating-voltage Ue:		max. 440 VAC	max. 690 VAC
rated voltage Un:		440 VAC	690 VAC
rated insulation voltage Ui:		440 VAC	690 VAC
rated impulse withstand voltage Uimp		4 kV	6 kV

6,0 bis 28,0A

Type label:

Client:	# # #
Series no.:	# # #
Type designation:	MIP BR4000
Material no.:	# # #
Type of electricl system:	TN-S
Rated Voltage Un:	415V AC
Rated Frequency fn:	50 Hz
Rated current InA:	# # #
Rated short current ICC:	25 kA
Control Voltage:	24V DC
Protection class:	IP 54
Year of manufacturing:	# # #
Listed standards:	IEC 61439-1 IEC 61439-2 DIN EN 60204-1

0,16 - 0,76

50 Hz

III

3

A

IP 54

0 °C

40 °C

2000 m

24 V

24 V

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TECHNICAL DATA AND CONSTRUCTION DETAILS

IEC 61355

Full designation		Structure description
mounting-location		
+BSF	Baseframe	
+ENG	Engine	
+GEN	Generator incl. terminal box	
+MIP	MTU Genset interface panel	
+MMC	MTU Genset control	
+GCB	Generator circuit breaker	
+MCB	Mains circuit breaker	
+FAN	Fan control	
+TNK	Fuel tank	
+EXT	Devices and enclosures outside any other enclosure, not delivered by MTU	
+CUS	Customer page	
+DOK	documents	
+TB1	Alternator terminal box	

Full designation		Structure description
document-type		
&EAA	Electrical cover sheet	
&EAB	Electrical table of contents	
&EDB	Description	
&EFS	Wiring diagram	
&ETL	Layout	
&EMA	Electrical terminal diagram	
&EMB	Electrical cable plan	
&EPC	Bill of material	

Full designation		Structure description
system		
=EKW	Electrical power	
=CON	Control voltages	
=EES	Engine emergency stop	
=PLC	Programmable logic controller	
=COM	Communication	
=STR	Engine starting	
=MCC	Mixture cooling circuit	
=ECC	Engine/jacket water circuit	
=HWC	Heating water circuit	
=FUL	Fuel supply	

TECHNICAL DATA AND CONSTRUCTION DETAILS

Wire colors within enclosure			
According IEC EN 60204-1			
Rated voltage	Above 48VAC	black	
Neutral wire	N	light blue	
Protective earth	PE	green/yellow	
Control voltage	24-240V AC	red	
	0V AC	red/white	
	24-110V DC	dark blue	
PLC-Controller	0V DC	grey	
		white	
		grey	
Measuring technique			
wires before Mainswitch		orange	

		L1 - L2 - L3 1L1 - 1L2 - 1L3 N PE
Main power before main switch		
Main power main switch on		
Neutral wire		
Protective conductor		
Mains power supply	400VAC	-XD00
Power supply Heating GEN	230VAC	-XD01
Power supply heatg. cool. water	400VAC	-XD02
Emergency stop		-XD06
Supply DC	24VDC	-XDL+
Supply DC	0VDC	-XDL-
Harness W011-ECU9 Digital		-1XD01
Harness W011-ECU9 Analog		-1XD02
Harness W020 Engine		-2XD01
Harness W030 Generator signal		-3XD01/-3XD02
Harness W031 Generator PT100		-3XD03
Option		
Customer interface WD04.1 / WG04.2 / WG04.3		-4XD01/-4XD02/ -4XD03
Harness W050 Power panel		-5XD01
Control cable WD06.1 Generator circuit breaker		-6XD01
Control cable WD07.1 Mains circuit breaker		-7XD01
Control cable WD08.1 Fan control enclosure		-8XD01
Control cable WD09.1/WD09.2 Fuel control		-9XD01/-9XD02
CAN bus		-XF01

Tightening torque			
Power relay	75A	M3,5x5 M5x8	1,1-1,2 Nm 3,2-3,5 Nm
Starter	Prestolite 9KW	M5 M12	2,0-3,0 Nm 27-33 Nm
Starter	Prestolite 15KW	M5 M10	1,5-1,7 Nm 9-10 Nm
Starter	Delco 9KW	Housing 10-32 INCH 1/2-13 INCH	18-23 Nm 1,8-3,4 Nm 27-34 Nm
Battery disconnect switches		M10 M12	15-20 Nm 18-22 Nm
Battery terminals	DIN 72331	Cable M6 M8	5-6 Nm Class 4,6 / 3,5 Nm Class 4,6 / 8,4 Nm
Grounding		M8 M10 M12 M14	Class 8,8 / 28 Nm Class 8,8 / 54 Nm Class 8,8 / 93 Nm Class 8,8 / 148 Nm